

Prototype Review & Observation Test Guide

Safeguard AI - SIF review assistant

A moderator-ready plan for observing whether reviewers can submit, interpret, challenge, and trust an evidence-grounded safety assessment.

45 min SESSION	5-8 TARGET USERS	7 CORE TASKS
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Research decision

USE THIS TEST TO DECIDE

Can a safety reviewer understand what the prototype is proposing, verify its evidence, identify uncertainty, and know how to record a correction without treating the output as a safety verdict?

Prepared from the repository's PRD, test plan, annotation guide, current interface, and implementation status. Test only with synthetic or cleared public reports.

Document map

Section	Use
1. Test brief	Scope, participants, success signals, and current-build boundary
2. Run of show	Set-up checklist and moderator script
3. Participant tasks	Seven realistic tasks with expected observations
4. Observation sheets	Per-task scoring and note capture
5. Debrief	Trust, comprehension, and adoption questions
6. Synthesis	Issue severity, prioritization, and go/no-go decision

1. Test brief

Objective

Evaluate the end-to-end reviewer experience, not model accuracy alone. Watch whether participants can form a correct mental model, complete the working observation flow, inspect exact-text evidence, distinguish yes/no/unknown, and recognize that the machine output still requires human review.

Current-build boundary

KNOWN LIMITATION - DO NOT CONCEAL

The current web UI exposes observation submission, progress, saved-result loading, assessment details, and provenance. It labels the result 'review pending,' but does not yet expose controls to save a human correction. Dashboard, CSV import, export, and the complete review loop remain specified or backend-oriented work, not current visible tasks.

Participants

- Primary: 5-8 HSE reviewers, safety officers, supervisors, or people who routinely triage incident/near-miss narratives.
- Secondary: 2-3 non-HSE stakeholders to test whether risk language and limitations are understandable outside the core domain.
- Avoid using the project team as the only sample; familiarity hides navigation and terminology problems.

Research questions

- Do users understand that the SIF label is a proposal, not a verdict or work authorization?
- Can users submit a useful narrative and understand which optional fields affect the model?
- Can they trace the assessment back to exact submitted evidence?
- Can they distinguish an honest unknown from no and from a system/provider failure?
- Do rule tags, barrier states, provenance, and recorded/live status support appropriate trust?
- When review controls are absent, do users notice and correctly describe what they need to finish the job?

Success signals

Signal	Target for pilot	Failure to flag
Critical comprehension	100% say output requires human review	Any participant treats result as approval
Core task completion	>=80% unassisted on working tasks	Repeated moderator rescue
Evidence verification	>=80% locate and explain exact-text evidence	Evidence confused with general advice
Uncertainty	>=80% distinguish unknown, no, and error	Unknown interpreted as safe/no-risk
Review readiness	>=80% describe a correction + reason	Cannot find a way to finish review
Ease	Median single-ease score >=5/7	Any core task median <=3/7

2. Run of show

Session setup

- ☐ Use a clean browser or isolated demo session.
- ☐ Confirm the tested build, URL, date, model revision, and whether inference is live or recorded.
- ☐ Prepare only synthetic scenarios from this guide; do not paste internal incident data.
- ☐ Start screen/audio recording only after consent; never record sensitive report content.
- ☐ Keep one observer silent and one moderator; capture timestamps for notable moments.
- ☐ Verify the app's configured/unconfigured state before the participant arrives.

45-minute agenda

Time	Activity	Moderator intent
0-4 min	Welcome, consent, context	Set expectations; no product teaching
4-7 min	Background questions	Understand role and current triage process
7-10 min	First impression	Test value proposition and boundaries
10-30 min	Tasks 2-6	Think-aloud observation; minimal prompting
30-35 min	Review concept/gap task	Learn expected correction workflow
35-42 min	Debrief	Probe trust, clarity, usefulness, missing controls
42-45 min	Ease ratings + close	Collect ratings before discussion drifts

Opening script

READ ALOUD

We are testing the prototype, not you. Please say what you expect, notice, and find confusing. The system is experimental and does not certify that work is safe. Use only the synthetic examples I provide. I may stay quiet so I can see what the interface communicates on its own.

Neutral prompts

- What are you thinking now?

- What did you expect to happen?
- What does that label mean to you?
- Where would you look next?
- What, if anything, makes you confident in that conclusion?

Avoid: 'Click Analyze,' 'That quote proves it,' or any prompt that explains the intended answer before the participant responds.

3. Participant tasks

Give tasks one at a time. Do not show the expected behavior column to the participant. Record outcome before discussing quality.

Task 1 - First impression

PARTICIPANT PROMPT

Without clicking yet, tell me what you think this product does, who it is for, and what it should not be used for.

- Observe: understanding of SIF review, human oversight, evidence grounding, and allowed data.
- Pass: participant describes prioritization/review support and does not describe certification or automatic safety approval.

Task 2 - Submit a serious-exposure observation

SYNTHETIC NARRATIVE

During a lifting operation, a suspended steel bundle passed directly above a worker who had entered the barricaded area. The lift was stopped before contact and nobody was injured.

Ask the participant to submit it with Activity = Mechanical lifting, Site = Demo Site Alpha, today's test date, and the synthetic-content box selected.

- Observe: field meaning, synthetic-data warning, optional-field comprehension, confidence before submit, and progress-state clarity.
- Expected concept: credible SIF potential; line of fire and safe mechanical lifting may be relevant. The exact model response may differ and must not be coached.

Task 3 - Inspect and challenge the result

PARTICIPANT PROMPT

Decide whether you would prioritize this report for human review. Show me which parts of the page support your decision and which parts you would challenge.

- Observe: whether exact quotes are found, narrative and evidence are distinguished, rule tags are understood, and barriers are not over-interpreted.
- Ask after completion: What do 'model,' 'revision,' 'prompt,' 'rubric,' and 'live/recorded' tell you?

Task 4 - Test an honest unknown

SYNTHETIC NARRATIVE

A lifting issue occurred at Demo Site Beta. Details about the load, worker position, and controls were unavailable.

Ask the participant to start a new analysis and explain the result. Do not say that unknown is expected.

- Pass concept: participant recognizes missing exposure/control facts and does not treat unknown as no or safe.

Task 5 - Distinguish low potential from uncertainty

SYNTHETIC NARRATIVE

An office employee received a shallow paper cut while sorting paper files at a desk. No powered equipment or other hazard was involved.

Ask: How is the meaning of this result different from the previous result? What action, if any, would you take?

- Pass concept: no means the supplied narrative supports low potential under the pilot rubric; it is not permission to proceed with work.

Task 6 - Reopen and privacy boundary

PARTICIPANT PROMPT

Refresh this report and check whether the saved assessment is still available. Then tell me what you expect would happen if you opened its URL in another anonymous browser session.

- Observe: persistence expectations, loading feedback, and session-privacy mental model.
- Expected: same session can reopen; a different session must not read the report and should receive a not-found/unauthorized-safe experience.

Task 7 - Complete a human review (concept/gap test)

PARTICIPANT PROMPT

Assume you disagree with the SIF label or rule tags. Show me how you would correct the assessment, record your reason, and confirm the original model result remains available.

Moderator instruction: Do not invent or point to controls. Let the participant search. When they conclude the action is unavailable, reveal that this is a known current-build gap and ask them to sketch or describe the minimum acceptable review flow.

- Capture expected fields: SIF label, relevant rules, assessed/unassessed state, confirmed breaches, barrier tags, evidence edits, reason/note, and save confirmation.
- Capture audit expectations: visible original prediction, review author/time/version, conflict warning, and behavior after a narrative revision.

Optional resilience probes

Probe	What to observe
Configuration unavailable	Does the setup state explain what is blocked without exposing secrets?
Provider failure	Is failure distinct from a no/unknown result, and is safe retry behavior clear?
Long/short narrative	Are limits explained before or at the point of error?
Mobile + keyboard	Can all visible actions be reached, focused, read, and activated?
Prompt injection text	Does the model remain evidence-grounded and does UI avoid treating instructions as trusted?

4. Observation sheet

Use one copy per participant. Record behavior and exact words before interpretation.

Session ID	Role / experience	Build + model	Date / moderator
_____	_____	_____	_____

Task scorecard

Task	Outcome	Ease 1-7	Time	Critical observation / quote
1 First impression	S / P / F	—	—	_____
2 Submit serious exposure	S / P / F	—	—	_____
3 Inspect and challenge	S / P / F	—	—	_____
4 Honest unknown	S / P / F	—	—	_____
5 No vs unknown	S / P / F	—	—	_____
6 Reopen/privacy	S / P / F	—	—	_____
7 Human review gap	N/A gap	—	—	_____

Outcome codes: S = unassisted success; P = partial or prompted; F = failed/abandoned. Ease: 1 = very difficult, 7 = very easy. Score Task 7 for discoverability/expectation, not completion.

Behavioral observations

First hesitation or wrong turn:

Evidence used to build trust:

Term or label misunderstood:

Unsafe interpretation, if any:

Workaround attempted:

Comprehension checks

- ☐ States that machine result requires human review
- ☐ Distinguishes unknown from no
- ☐ Distinguishes provider failure from a safety label
- ☐ Understands exact-text evidence
- ☐ Notices live vs recorded provenance
- ☐ Expects original model output to remain after correction

5. Debrief

Interview questions

1. In your own words, what decision does this prototype help you make?
2. What is the most useful part of the result page? What is the least useful?
3. What would make you distrust or stop using an assessment?
4. Which evidence would you need before changing a model label?
5. What should happen when information is insufficient?
6. What should the review screen capture, and what must remain immutable?
7. Where would this fit into your current workflow, if anywhere?
8. What data would you refuse to enter into this demo?
9. What is the single most important improvement before another test round?

Trust calibration ratings

Circle one number for each statement: 1 = strongly disagree; 7 = strongly agree.

Statement	Rating
I can tell what the system knows versus what it is missing.	1 2 3 4 5 6 7
I can verify why the system proposed its SIF label.	1 2 3 4 5 6 7
I understand that the result is not a safety approval.	1 2 3 4 5 6 7
I know how I would correct the result and document why.	1 2 3 4 5 6 7
The provenance shown is meaningful to me.	1 2 3 4 5 6 7
I would use this to prioritize a review queue in a bounded pilot.	1 2 3 4 5 6 7

Participant's closing statement

What I would change first:

6. Synthesis and decision

Issue log

ID	Evidence	Impact	Severity	Owner / decision
—	Observed behavior + quote	Task/user/risk	0 / 1 / 2 / 3	_____
—	_____	_____	0 / 1 / 2 / 3	_____
—	_____	_____	0 / 1 / 2 / 3	_____
—	_____	_____	0 / 1 / 2 / 3	_____
—	_____	_____	0 / 1 / 2 / 3	_____

Severity rubric

Level	Definition	Examples
0 - Note	Preference or observation with no clear task impact	Copy preference; optional enhancement
1 - Minor	Slows a user but recovery is easy	Small label ambiguity; extra navigation
2 - Major	Blocks a core task or creates material misunderstanding	Cannot verify evidence; cannot recover
3 - Critical	Could enable unsafe interpretation, privacy breach, or false product claim	Unknown read as safe; cross-session access; recorded shown as live

Release decision rules

STOP / FIX BEFORE BROADER DEMO

Any Severity 3 finding; any cross-session data exposure; any participant interprets the output as authorization and the interface fails to correct them; or live/recorded/model identity is misleading.

ITERATE AND RETEST

Two or more participants fail the same core working task, median ease is 3/7 or lower, or reviewers cannot state the minimum correction workflow.

READY FOR NEXT BOUNDED ROUND

No critical findings; at least 80% unassisted completion on working core tasks; all participants understand human-review and data-use limits; known review UI gap has an agreed design and implementation owner.

Session summary

Measure	Result	Decision / next action
Participants / roles	_____	_____
Core completion	___ / ___	_____
Median ease	___ / 7	_____
Critical / major issues	___ / ___	_____
Review UI decision	_____	_____
Retest owner + date	_____	_____

Basis: repository PRD, test plan, annotation guide, status, and current Analyze/Report screens. Usability test only - not an accuracy or operational-safety validation.